

Pramil Paudel

Machine Learning, Data Science, Realtime Processing, Privacy Preservation, and Web Applications

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SUMMARY

Ph.D. researcher in computer science with 10+ years of combined academic and industry experience in privacy-preserving AI, lensless imaging, and real-time physics-aware ML systems. Specializes in optics-driven, reconstruction-free visual intelligence and previously developed enterprise web solutions for the U.S. healthcare and insurance sector.

SKILLS

Programming	Python; C/C++; MATLAB; Java
Tools	PyTorch; SciPy; OpenCV; Tensorflow; Git; Linux; ROOT; CUDA
ML/AI	CNNs; Transformers; GANs; Transfer Learning; VGG; ResNet
Domains	Lensless Imaging; Steganography; Privacy; Remote Sensing; High-Energy Physics

PROFESSIONAL ACADEMIC EXPERIENCE

Graduate Research Assistant / Graduate Teaching Assistant | University of Kansas Lawrence, KS 2020–Present
Graduate Research Assistant (GRA)

- *PPLiC — Privacy-Preserving Lensless Image Classification*
 - Developed a robust classification pipeline resistant to machine learning-based adversarial attacks.
 - Advanced lensless imaging as a secure alternative for privacy-aware visual sensing and communication.
- *RFLiS — Reconstruction-Free Lensless Image Recognition*
 - Designed a state-of-the-art lensless image classifier that bypasses spatial images reconstruction entirely.
 - Demonstrated feasibility for surveillance and UAV-based remote sensing under high data-volume constraints.
- *Quantum Tomography (QT)–Inspired Machine Learning*
 - Investigating dimensionality reduction and real-time processing on high-dimensional data.
 - Developed a prototype classifier leveraging quantum tomography principles for real-time inference.
- *Instructor — KU Center for Digital Inclusion*
 - Designed and taught a hands-on “Introduction to Computing” course for justice-involved individuals.
 - Delivered an applied “Python for Problem Solving” module focused on digital reintegration and empowerment.

Graduate Teaching Assistant (GTA)

- Courses Taught & Responsibilities
 - Assisted in teaching Software Engineering II (Capstone), Digital Logic, Analysis of Algorithms, and Object-Oriented Programming.
 - Led lab sessions, advised senior capstone teams, delivered instructional support, and evaluated student assignments and projects.

PROFESSIONAL INDUSTRIAL EXPERIENCE

Instructor & Technical Trainer | Four G Engineering Kathmandu, Nepal Sept 2017–Dec 2019

- Taught microcontrollers, embedded systems, and full-stack Java web development.
- Designed hands-on labs and industry-aligned technical training modules.

Solution Architect | Renegade Insurance Remote / USA Dec 2018–Dec 2019

- Architected a resilient Java-based platform for real-time policy generation.
- Implemented CI/CD pipelines and RESTful microservices for insurance carrier integrations.

Freelance Software Engineer | Upwork / Freelancer Remote Sept 2017–Dec 2019

- Built REST APIs using Groovy/Grails and implemented Elasticsearch-based ad search engines.
- Prototyped autonomous IoT microservices in Java for pilot deployments.

Engineering Manager | National Incubation & Research Center Kathmandu, Nepal May 2017–Sept 2017

- Led a 20+ engineer team and deployed Redmine, GitLab, and Jenkins CI/CD infrastructure.
- Designed AWS cloud environments and Spark-based distributed data processing jobs.

Lead Software Engineer | Cedargate Inc. (formerly Deerwalk) Kathmandu, Nepal Dec 2014–May 2017

- Built large-scale healthcare claims processing pipelines using Hadoop/Spark and Scala.
- Delivered production-grade cloud analytics (ETL → reporting pipelines) for U.S. clients.

Software Developer | Yotcha Inc. Remote / Singapore

Jan 2015 – Jan 2016

- Developed full-stack real-estate web applications using Groovy on Grails.
- Delivered both frontend user-facing features and backend data processing workflows.

CONFERENCES AND PRESENTATIONS

Invited Presentations / Workshops

- **Version Control, Shell Scripting, and Remote Computing for Academic Research**
Bootcamp: Computational Tools for Physics • indico.global/event/14047

Conference Presentations

- **ICMLA 2025 (Scheduled): RFLiS — Reconstruction-Free Lensless Image Classification**
International Conference on Machine Learning and Applications (ICMLA) • icmla-conference.org/icmla25
- **AI for the Electron-Ion Collider (EIC)(Scheduled) —TBD**
A.I. for the Electron-Ion Collider Program • aic.ai
- **Robust Privacy-Preserving Classification for Lensless Images**
17th Central Area Networking and Security (CANSec) Workshop • sites.google.com/view/cansec2024
- **Electron Compton Polarimetry R&D**
Summer 2024 Joint EICUG / ePIC Collaboration Meeting • indico.bnl.gov/event/20727

Official Designer

- **UPC 2023: International Workshop on the Physics of Ultra-Peripheral Collisions**
CERN (Indico) • indico.cern.ch/event/1263865

EDUCATION

University of Kansas (KU)

Ph.D. in Computer Science (*in progress*)

Jan 2020 – Present

Tribhuvan University

B.E. in Electronics & Communication Engineering

Dec 2010 – Dec 2014

PUBLICATIONS (SELECTED)

- S. Bhattacharai, **P. Paudel**, Z. Li, B. Luo, and F. Li. *Robust Privacy-Preserving Classification for Lensless Images*. IEEE MASS 2024, pp. 371–379. <https://doi.org/10.1109/mass62177.2024.00056>
- **P. Paudel**, F. Li. *RFLiS: Reconstruction-Free Lensless Image Classification*. Accepted at IEEE ICMLA 2025. (to appear)
- **Paudel, P.**, B. Adhikari (2017). *Basic Electronics Engineering (Textbook)*. Heritage Publishers & Distributors Pvt. Ltd., Kathmandu, Nepal. ISBN 978-9937-7031-1-6 • Link: Title and Prefaces of the Book

Additional note: Engaged in multiple collaborative research efforts with the KU ALICE High Energy Physics Lab (CERN). Full publication and citation record available via [Google Scholar Link](#)

OUTREACH AND SOCIAL WORK

Secretary, International Nepali Literature Society (INLS), Kansas2024–2026

President, Nepalese Student Association, University of Kansas2021–2022

President, Engineering Literature Forum2012–2014

Newspaper Columnist, Samadhan National Daily2013–2014

Authored literary articles, poems, and public presentations featured in national newspapers and cultural forums in Nepal • [Link to the Blog](#)

AWARDS

Graduate Student Travel Fund, University of Kansas2025

CANsec Conference Travel Award2022,2024,2025

Lydar Fellowship, University of Kansas2020

Parkins EECS Scholarship, University of Kansas2020